

Bayesian Additive Regression Trees for Causal Inference

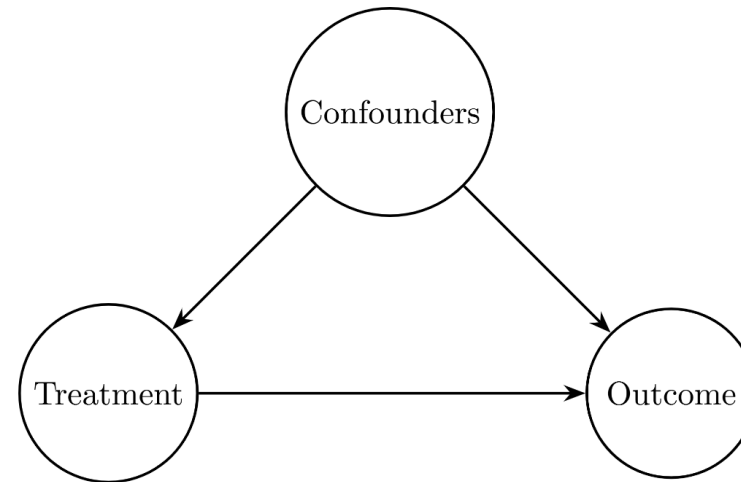
An Empirical Evaluation

Luise Sessler

13.07.2026

Motivation

- Objective: Estimate the causal effect of a treatment on an outcome
- Observational data are affected by confounding variables



- ➔ Flexible methods are needed to model complex relationships
- ➔ Potential approach: Bayesian Additive Regression Trees (BART)

Research Questions

1. How do different BART-based estimators perform relative to each other in estimating treatment effects?
2. How do BART-based estimators perform compared with established methods for causal effect estimation?

Potential Outcomes Framework (Rubin, 1974)

- Causal effect: difference between two potential outcomes
- For individual i :

$$\tau_i = Y_i(1) - Y_i(0)$$

where

$Y_i(1)$: Outcome if individual i receives treatment

$Y_i(0)$: Outcome if individual i does not receive treatment

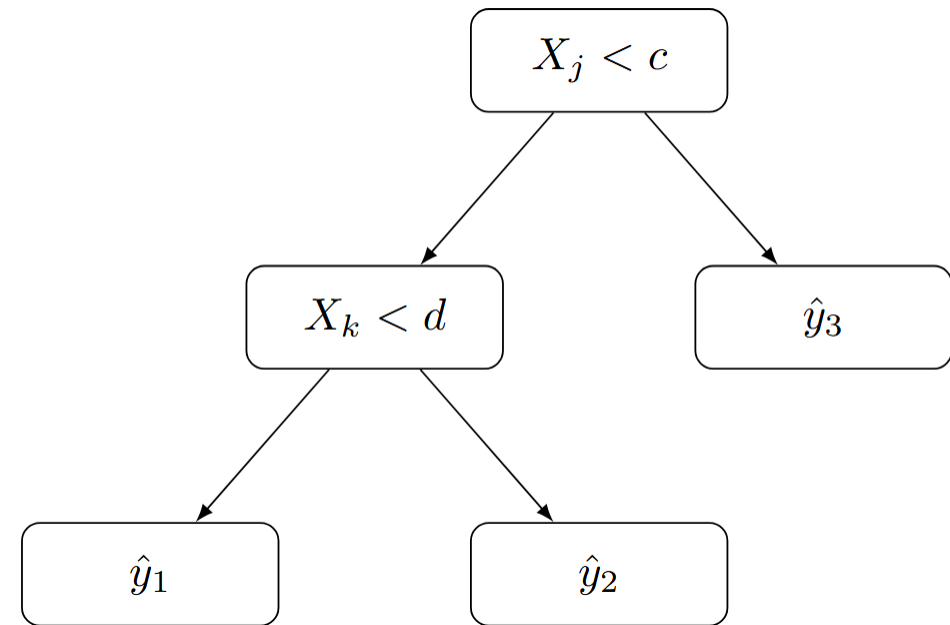
- Average Treatment Effect (ATE):

$$\tau = \mathbb{E}[Y(1) - Y(0)]$$

- Fundamental problem of causal inference: Only one potential outcome is observed for each individual.

Bayesian Additive Regression Trees (Chipman et al., 2011)

- Ensemble of weak regression trees
- Bayesian framework provides regularization and uncertainty estimates
- Captures nonlinearities and interactions automatically



BART for Causal Effect Estimation

- Idea: Estimate the missing potential outcomes (Hill, 2011)
- Approaches:
 - S-learner: One BART model which includes treatment as an additional covariate (Hill, 2011)
 - T-learner: Two separate BART models for the treatment and the control group (Künzel et al., 2019)
 - PS-BART: S-learner which includes propensity scores $e(X_i)$ as an additional covariate (Hahn et al., 2020)

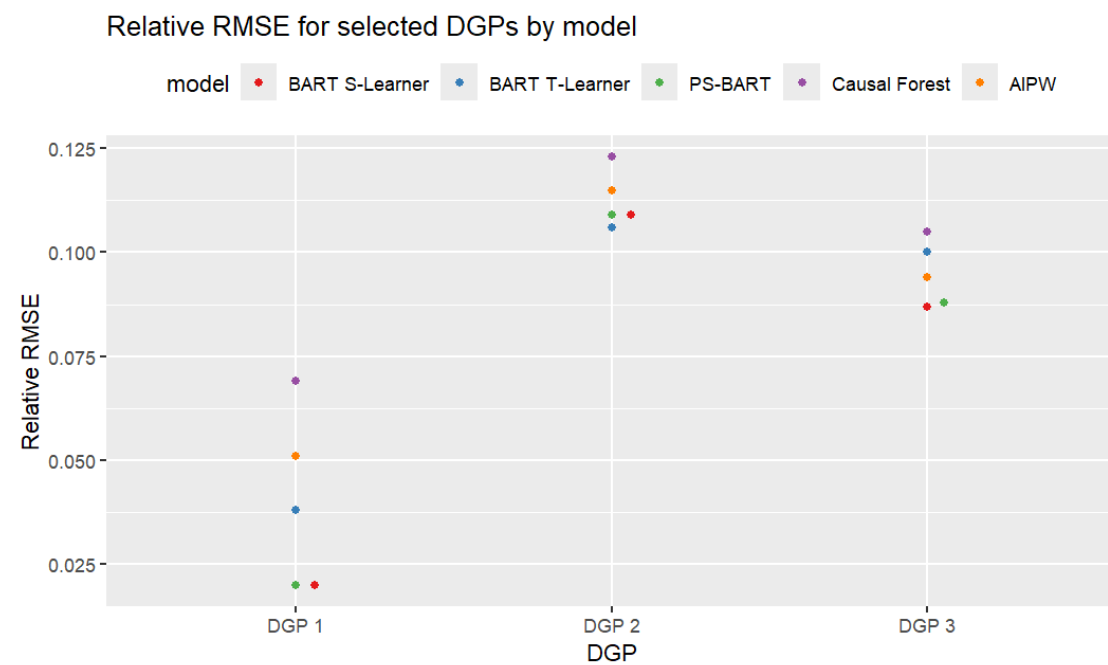
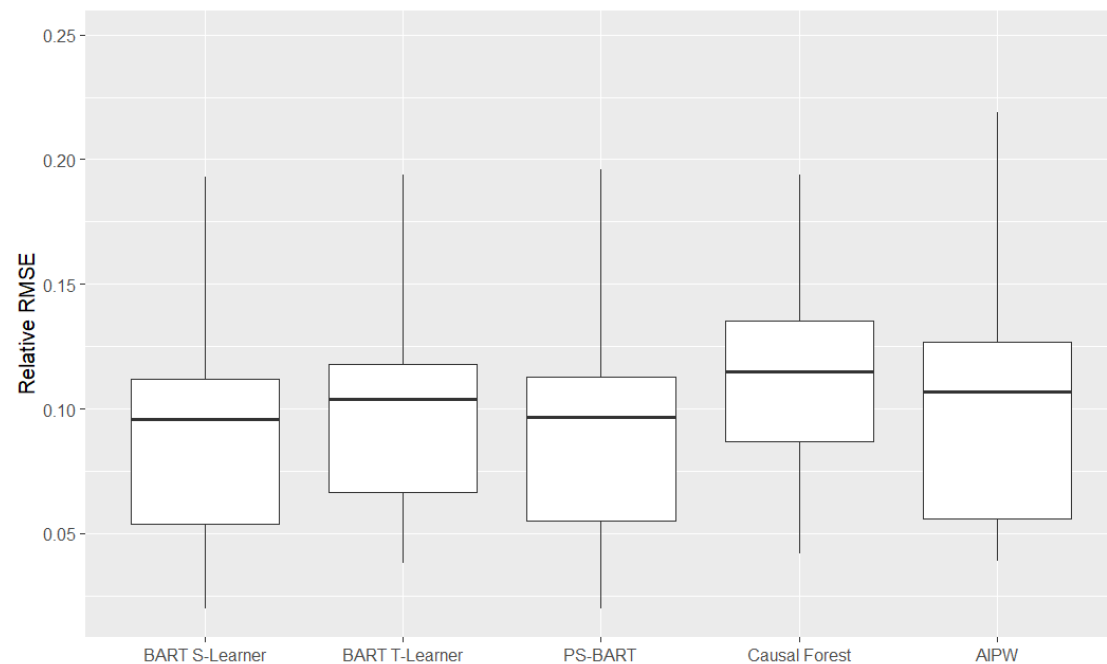
$$e(X_i) = P(W_i = 1 \mid X_i)$$

(Rosenbaum et al., 1983)

Empirical Evaluation

- BART-based estimators:
 - BART S-learner
 - BART T-learner
 - PS-BART (Propensity scores estimated using logistic regression)
- Benchmark estimators:
 - Causal Forest (Wager & Athey, 2018)
 - Augmented Inverse Probability Weighting (AIPW) (Robins et al., 1994)
- Experimental Setup:
 - 16 synthetic and semi-synthetic datasets
 - Estimation of the Average Treatment Effects
 - Evaluation metric: Root Mean Squared Error (RMSE)

Results



Conclusion

- BART-based approaches generally outperformed benchmark estimators.
- The S-learner performed better than the T-learner in most scenarios.
- Including propensity scores did not substantially improve performance.
- No single estimator performed best across all scenarios.
- Further research is needed to identify the data characteristics under which different estimators perform best.

Literature

- Chipman, H. A., George, E. I., & McCulloch, R. E. (2010). BART: Bayesian additive regression trees. *The Annals of Applied Statistics*, 4(1), 266–298. doi: 10.1214/09-AOAS285
- Glynn AN, Quinn KM. An Introduction to the Augmented Inverse Propensity Weighted Estimator. *Political Analysis*. 2010;18(1):36-56. doi: 10.1093/pan/mpp036
- Hahn, P. R., Murray, J. S., & Carvalho, C. M. (2020). Bayesian regression tree models for causal inference: Regularization, confounding, and heterogeneous effects (with discussion). *Bayesian Analysis*, 15(3), 965-1056.
- Hill, J. L. (2011). Bayesian Nonparametric Modeling for Causal Inference. *Journal of Computational and Graphical Statistics*, 20(1), 217–240. doi: 10.1198/jcgs.2010.08162
- Künzel, S. R., Sekhon, J. S., Bickel, P. J., & Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the national academy of sciences*, 116(10), 4156-4165. doi: 10.1073/pnas.1804597116
- Robins, J. M., Rotnitzky, A., & Zhao, L. P. (1994). Estimation of Regression Coefficients When Some Regressors are not Always Observed. *Journal of the American Statistical Association*, 89(427), 846–866. doi: 10.1080/01621459.1994.10476818
- Rosenbaum, Paul R., and Donald B. Rubin. 1983. The central role of the propensity score in observational studies for causal effects. *Biometrika* 70:41–55.
- Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5), 688–701. doi: 10.1037/h0037350
- Wager, S., & Athey, S. (2018). Estimation and Inference of Heterogeneous Treatment Effects using Random Forests. *Journal of the American Statistical Association*, 113(523), 1228–1242. doi: 10.1080/01621459.2017.1319839